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Efficacy and safety of osteopathic manipulative treatment: an overview of systematic reviews

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ABSTRACT

Objective To summarise the available clinical evidence on the efficacy and safety of osteopathic manipulative treatment (OMT) for different conditions.

Design Overview of systematic reviews (SRs) and meta-analyses (MAs). PROSPERO CRD42020170983.

Data sources An electronic search was performed using seven databases: PubMed, EMBASE, CINAHL, Scopus, JBI, Prospero and Cochrane Library, from their inception until November 2021.

Eligibility criteria for selecting studies SRs and MAs of randomised controlled trials evaluating the efficacy and safety of OMT for any condition were included.

Data extraction and synthesis The data were independently extracted by two authors. The AMSTAR-2 tool was used to assess the methodological quality of the SRs and MAs. The overview was conducted and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement.

Results The literature search revealed nine SRs or MAs conducted between 2013 and 2020 with 55 primary trials involving 3740 participants. The SRs reported a wide range of conditions including acute and chronic non-specific low back pain (NSLBP, four SRs), chronic non-specific neck pain (CNSNP, one SR), chronic non-cancer pain (CNCP, one SR), paediatric (one SR), neurological (primary headache, one SR) and irritable bowel syndrome (IBS, one SR). Although with a different effect size and quality of evidence, MAs reported that OMT is more effective than comparators in reducing pain and improving functional status in acute/chronic NSLBP, CNSNP and CNCP. Due to small sample size, presence of conflicting results and high heterogeneity, questionable evidence existed on OMT efficacy for paediatric conditions, primary headache and IBS.

No adverse events were reported in most SRs. According to AMSTAR-2, the methodological quality of the included SRs was rated low or critically low.

Conclusion Based on the currently available SRs and MAs, promising evidence suggests the possible effectiveness of OMT for musculoskeletal disorders. Limited and inconclusive evidence occurs for paediatric conditions, primary headache and IBS. Further well-conducted SRs and MAs are needed to confirm and extend the efficacy and safety of OMT.

Strengths and limitations of this study

- This systematic overview included a comprehensive literature search for evidence on the efficacy and safety of osteopathic manipulative treatment (OMT) for any condition.
- The present overview was conducted according to the Cochrane Handbook for the Systematic Review of Interventions and reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.
- The inclusion criteria were restricted to systematic reviews and meta-analyses of randomised controlled trials that included patients with any conditions.
- Since only randomised controlled trials in which OMT was performed by osteopathic physicians or osteopaths were included, some relevant systematic reviews could have been missed.
- The quality of the evidence from the included systematic reviews and meta-analyses was assessed according to the AMSTAR-2 tool.

INTRODUCTION

Osteopathic medicine, depending on different legal and regulatory structures around the world, is a medical profession (eg, USA), an allied health profession (eg, UK) or a part of complementary and alternative medicine (CAM; eg, Italy or France), developed by Andrew Taylor Still in the late 1800s in the Midwestern USA.[1] This therapy is based on the principle that the structure (anatomy) and function (physiology) of the individual's body are closely integrated and that a person's well-being depends on the balance of neurological, musculoskeletal and visceral structures.[1]

Osteopathic medicine is provided on almost every continent, and in 2020, a survey estimated that 196 861 osteopathic practitioners provide osteopathic care worldwide in 46 countries.[2]

Osteopathic medicine plays an important role primarily in musculoskeletal healthcare. A recent survey conducted in Switzerland[3] on a sample of 1144 patients showed that over 80% of patients had requested an osteopathic consultation for musculoskeletal pain [mainly

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low back pain (LBP), neck pain and headaches]. Similar results were reported by a survey conducted in the UK on a sample of approximately 1600 patients with pain in the lumbar spine, cervical spine and pelvic region.[4] Finally, a prospective study on 14 000 patients in Quebec, Canada reported musculoskeletal pain, localised in the spine, thorax, pelvis and limbs as the most common reason for osteopathic consultations.[5]

Osteopathic manipulative treatment (OMT) is defined in the Glossary of Osteopathic Terminology as 'the therapeutic application of manually guided forces by an osteopathic practitioner to improve physiologic function and/or support homeostasis that has been altered by somatic dysfunction'.[6] OMT refers to a number of various types of approaches and techniques such as myofascial release, mobilisation, osteopathy in cranial field and visceral manipulation, in order to optimise the body's normal self-regulating mechanisms. The aim of OMT is to solve somatic dysfunction (International Classification of Diseases 10 CM diagnosis code M99.00–09), although other care aspects have been proposed.[1 7]

In recent years, a number of systematic reviews (SRs) and meta-analyses (MAs) have been published to evaluate the clinical efficacy and safety of osteopathic medicine for conditions such as LBP, neck pain and migraine. However, due to differences in methodologies and the quality of SRs, no clear conclusions were achieved. The aim of this overview is to summarise the available clinical evidence on the efficacy and safety of OMT for different conditions. This overview may be relevant to clinicians and policymakers to better understand in which conditions osteopathic medicine can be an effective and safe complementary therapy.

METHODS

The overview was conducted according to the Cochrane Handbook for the Systematic Review of Interventions (Cochrane Book) and reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[8–10] The protocol of the overview has been registered on PROSPERO (CRD42020170983).

Patient and public involvement statement

For this overview of SRs and MAs, patients or the public were not involved.

Eligibility criteria

Type of review

This overview included only SRs and MAs, published as a full paper, of randomised controlled trials (RCTs), which are well known to be the gold standard for evaluating the efficacy of an intervention.[11] SRs evaluating the inter-rater or intrarater reliability for any type of osteopathic approach were excluded. SRs and MAs evaluating both RCTs and controlled clinical trials were excluded if a subanalysis for RCTs was not performed. SRs that did not meet all eligibility overview criteria were excluded. For SRs in which criteria were not understandable, the primary studies were analysed.

Participants/population

Participants were human, of any gender, age or clinical condition undergoing OMT. Reviews including osteopathic manipulation on animal models as well as on healthy volunteers were excluded.

Intervention

The intervention consists of OMT performed by osteopaths, osteopathic physicians or osteopathic trainees who used a black box method or a specific protocol without any restriction of approach and technique based on manual assessment, diagnosis and treatment in accordance with the osteopathic principle.[1 2] SRs, including primary studies on both OMT and other complementary manual interventions, were excluded if a subanalysis was not independently performed for each manual treatment. To verify that, osteopathic treatment was performed by osteopaths, the primary clinical trials were analysed.

Comparison

In order to retrieve all clinical evidence currently reviewed in SRs and MAs, the comparison group included placebo, sham OMT, light touch therapy, no treatment, waiting list, conventional treatment, physiotherapy or other complementary medicine treatments.

Setting

SRs with trials performed in any health-related settings and/or health promotion centres were included.

Main outcomes

The main outcomes were any clinically relevant endpoint measures, depending on the clinical condition reported in the SRs.

Any adverse events caused by OMT were extracted. Other types of outcomes such as prevalence of somatic dysfunction and inter-rater or intrarater reliability for any type of osteopathic approach were excluded.

Search strategy

A systematic literature search was carried out independently by two reviewers (DB and DR) using the following electronic databases: MEDLINE (PubMed), Cumulative Index to Nursing and Allied Health Literature (CINAHL), Excerpta Medica database (EMBASE), Joanna Briggs Institute database of SRs and implementation reports (JBI), Scopus, Prospero and Cochrane Library, all from their inception until 13 November 2021. No language or date restrictions were applied. The search strategy was performed using the following search terms: osteopathic treatment, osteopathic medicine, osteopathic manipulation, review, systematic review and meta-analysis. The reference list of the included SRs and MAs, as well as narrative reviews, were widely perused for the identification of additional articles. Full details of the search

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strategy for PubMed are provided in the *Appendix* (online supplemental materials).

Data collection and analysis

Study selection

The selection was performed independently by two authors (DB and DR). All the retrieved articles were imported into V.1.19.8 of the Mendeley software, and duplicate publications were excluded. Potential eligible SRs and MAs were read in the abstract and full text and independently evaluated by the two authors for inclusion in the overview. SRs and MAs were excluded if they did not meet the inclusion criteria, first at the title and abstract level and then at the full-text level. Disagreements were resolved through discussion and consensus between the two review authors; if no agreement was reached, the third member of the review team (FB) was then consulted. Weighted kappa statistics were calculated to measure agreement between the authors.

Data extraction and management

Two authors (DB and FB) independently extracted data using an Excel spreadsheet. We collected the following information (where available) from SRs and MAs: first author, year of publication and country of the corresponding author, date assessed as up to date, condition treated, number of included studies and participants, gender distribution and age, osteopathic interventions and cointervention description and number of treatments, control description, outcome measures, time points reported, reporting adverse events, primary studies of quality assessment included in each SRs and MAs, Grading of Recommendations Assessment, Development and Evaluation (GRADE) results (see the 'Strategy for data synthesis' section for more details), MAs data, if any, and SRs main results. We reported the mean difference (MD) or standard mean difference (SMD), 95% confidence interval (CI) and results of any test of heterogeneity reported in the relevant meta-analysis. When not reported in the SRs, mean and SD for continuous variables as well as median, intequartile range (IQR) and range for discrete variables were calculated (eg, patient's age, gender).

Assessment of the methodological quality of included SRs and MAs

The methodological quality of the included SRs and MAs was assessed using the AMSTAR-2 tool, which is designed to generate an overall rating based on weaknesses of some critical domains (items 2, 4, 7, 9, 11, 13 and 15).[10] AMSTAR-2 classifies the overall confidence of the results into four levels: high (no or one non-critical weakness), moderate (more than one non-critical weakness), low (one critical flaw with or without non-critical weaknesses) and critically low (more than one critical flaw or without non-critical weaknesses).[12] The quality assessment was evaluated independently by two authors (DB and FB), with any disagreements resolved through discussion with the third author (DR). To provide a simple indication of the results for the reader, for each domain, we used a 'stop-light' indicator where green indicates 'yes', yellow indicates 'partial yes' and red indicates 'no'. Weighted kappa statistics were calculated to measure agreement between the authors.

Overlapping SRs

In accordance with recent guidelines,[13 14] we decided to count the primary studies present in more than one SR only once. When more than one SR (which investigated the same research question and used the same primary studies) was identified, only the latest one was selected if it used the most rigorous criteria (eg, followed the PRISMA criteria, used the more recent SR/MA guidelines) to evaluate the methodological quality of the studies.

Strategy for data synthesis

Due to the overlap of studies and heterogeneity between reviews with regards to outcome measures, a critical synthesis of the results was performed. The methodological quality of RCTs can be evaluated using several scores, including the Jadad score, the PEDro scale and the Cochrane risk of bias (RoB) tool for randomised trials. Different versions of RoB are available, which refer to different updates of the Cochrane Handbook for SRs of intervention.[15 16] Moreover, for musculoskeletal disorders, the Cochrane Back Review Group (CBRG) now named Cochrane Back and Neck (CBN) Group has developed a specific RoB guideline (also for this guideline, different versions are available).[17–19] Considering different judgements in our overview, when possible, we have reported results (judgements) according to the last version of the RoB tool.[19 20] In table 1, authors' judgements are reported, while our update judgements are reported in the text. Once meta-analysis was performed, we reported the data synthesis used in the meta-analysis: effect size (ES) and heterogeneity. ES was reported according to Cohen.[21] Briefly, a small effect was defined as MD less than 10% of the scale and SMD less than 0.50%, a medium effect as MD from 10% to 20% of the scale and SMD from 0.50% to 0.80%, and a large effect was defined as MD greater than 20% of the scale and SMD scores greater than 0.80%.[19] Concerning heterogeneity, the following thresholds were considered for the interpretation of the reported I^2 statistic index: (1) 0% to 40% might not be important, (2) 30% to 60% may represent moderate heterogeneity, (3) 50% to 90% may represent substantial heterogeneity, (4) 75% to 100% considerable heterogeneity.[20] We reported the GRADE results as rated by the SRs authors. According to the GRADE approach, the quality of evidence for each outcome (considering the RoB, imprecision, inconsistency of results, indirectness of evidence and publication bias) can fall into four categories: high-quality evidence (further research is very unlikely to change confidence in the estimated effect), moderate quality (further research is likely to have an important impact on confidence in the estimate of effect

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Table 1 Characteristics of the included systematic reviews and meta-analyses

First author, year, country	Date assessed as up to date	Conditions	Trials number, participants number	Gender distribution, age (years)	Intervention (co-intervention): description. Number of treatments (SD)	Control or comparison description	Outcomes assessed	Time points reported	Main results	Definition used to measure AEs* Reported AEs	AMSTAR-2
Musculoskeletal conditions: low back pain											
De Oliveira Meirelles, 2013,[24] Brazil	NR	CLBP, CLBP in pregnancy, LBP with menopausal symptoms, LBP in obese, LBP with sciatica.	5 RCTs, 278 adults. 1 CLBP, 1 CLBP in pregnancy, 1 LBP with menopausal symptoms, 1 LBP in obese, 1 LBP with sciatica.	Gender:85% female,15% male. Mean age 40 (from 4 RCTs).	OMT (UOBC, SE): OCF, ART, HVLA, MRT, MET, range of motion technique. Treatments: median 10 (7-10)‡	SUT, NT, SM, chemonucleolysis.	Pain: VAS, dichotomous pain, pain scale.	Treatment time: 12 weeks and 15 weeks (from 2 RCTs). Evaluation: 1, 3 and 6 months (from 1 RCT).	OMT improved LBP in comparison with no intervention (but not with SM).	NR	Critically low
Franke, 2014,[25] Australia	NR	ANSLBP, CNSLBP, NSLBP in pregnancy, NSLBP in PP.	15 RCTs, 1502 adults. 10 NSLBP, 3 NSLBP in pregnancy, 2 NSLBP in PP.	Gender: NR. Mean age 36 (from 13 RCTs).	OMT (UC, heat and PT, UOBC, SE): NR. Treatments: median 4 (4–6)‡	SUT, NT, SM, UC, PT, SWD.	Pain: VAS, NRS, MGPQ. Functional status: RMDQ, OPQ, ODI, LBP_DQ, Kinematic of thoracic/lumbar spine /pelvis during forward flexion, QBPDS.	Period: 2–9 weeks, 1–3- 6 months.	OMT was effective in pain and functional status in ANSLBP, CNSLBP, NSLBP in pregnant and NSLBP in PP.	NR Only 4 RCTs reported AEs. 2 RCTs reported minor AEs such as stiffness and tiredness; 1 RCT reported that 6% of patients had AEs (but not serious). 1 RCT reported that no AEs occurred.	Low
Franke, 2017,[26] Australia	NR	ANSLBP, CNSLBP and /or pelvic pain during pregnancy and PP.	8 RCTs, 850 adults. 5 LBP in pregnancy, 3 LBP in PP.	Gender: 100% female, Mean age 29.5	OMT (UOBC): NR. Treatments: Pregnancy median 7 (5.5–7). Postpartum median 4 (4–4.5)‡	SUT, NT, UC.	Pain: VAS, QVAS, FP. Functional status: RMDQ, QPP, QBPDS, PGPQ, OPQ.	Pregnancy: ranging from 3 to 9 weeks; follow-up 1 and 2 weeks. Postpartum: 6 weeks. Follow-up 2 weeks.	OMT significantly improved pain functional status in women with LBP during pregnancy and PP.	NR No serious AEs (from 3 RCTs†). 1 RCT reported occasional tiredness in some patients.	Low
Dal Farra, 2020,[27] Italy	Inception to April 2020	CNSLBP	6 RCTs¶,739 adults.	Gender: NPTC Mean age 46 (from 4 RCTs), median age 41 (29-51)‡	OMT (SE, UC): HVLA, MET, CST, MFR, MVMA. Treatments: range 5–10 sessions, median 6 (5-8)‡	SM, PT, SE.	Pain: VAS. Functional status: RMDQ, ODI, SF-36, EQ-5D, BDI.	Ranging from 2 weeks to 6 months. Follow-up: from 1 month to 1 year.	OMT significantly improved pain and functional status in CNSLBP in the short-term (but not in the long-term).	Frequency of adverse events and/or relative study withdrawals, and self-reported scales and questionnaires including quality of life and psychological function (eg, fear avoidance beliefs, catastrophising, pain-related fear); additional indicators considered were frequency of analgesic and/or NSAIDs use, economic impact or cost reduction and patient's care satisfaction. No AEs (from 5 RCTs). 1 RCT reported increased back muscle spasticity in a patient.	Low
Musculoskeletal conditions: neck pain											

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Table 1 Continued

First author, year, country	Date assessed as up to date	Conditions	Trials number, participants number	Gender distribution, age (years)	Intervention (co-intervention): description. Number of treatments (SD)	Control or comparison description	Outcomes assessed	Time points reported	Main results	Definition used to measure AEs* Reported AEs	AMSTAR-2
Franke, 2015,[28] Australia	NR	CNSNP	3 RCTs, 123 adults.	Gender: NR. Mean age 44.	OMT (SUT, UC): NR. Treatments: median 5 (5-6)‡	SM, PT.	Pain: VAS, NRS, NPPQ. Functional status: NDI, NQ.	Ranging from 6 to 11 weeks. Follow-up: 3 months (in 2 RCTs).	OMT significantly improved pain, but not functional status in CNSNP.	NR Only 1 RCT reported not serious AEs, such as tiredness on the day of treatment and short-term aggravation of symptoms in other 'familiar' regions, were noted.	Low
Musculoskeletal conditions: chronic non-cancer pain											
Rehman 2020,[29] Canada	NR starting date. Until July 2019	CNCP: Fibromyalgia, TMD, CNSLBP, CNSBP, CNSNP, CNP.	7 RCTs**, 759 adults. 1 Fibromyalgia, 1 TMD, 1 NSNP, 1 CNSBP, 2 CNSLBP, 1 CNSNP.	Gender: 60% female, 40% male. Mean age 52 (from 5 RCTs), range 23–54 (from 2 RCTs).	OMT (non-steroidal medications, anti-inflammatory, analgesics and/or muscle relaxants, UC, SE, lumbar supports, physical therapies and CAM): MET, MFR, HVLA, BLT, CST, JA, MT, ST, FPR. Treatments: NR.	SUT, SE, PT, SC, use of an oral appliance, hot and/or cold packs, TENS, SM, LT, ROM activities, LTP.	Pain: VAS. Disability: RMDQ. SF-36, QOL	Duration of trial or follow-up period: ranging from 42 to 168 days (1–6 months).	OMT, in comparison to SC, was significantly effective in reducing pain and increasing disability as well as in improving QoL.	NR	Low
Paediatric conditions											
Posadzki, 2013,[30] South Korea	Inception to November 2012	Paediatric conditions: CP, respiratory conditions, OM, musculoskeletal function, ADHD, prematurity, IC, CNLDO, DV.	17 RCTs, 887 neonates/infants (from 16 RCTs)§. 2 CP, 4 respiratory conditions, 3 OM, 3 musculoskeletal function, 1 ADHD, 1 prematurity, 1 IC, 1 CNLDO, 1 DV.	Gender: NR. Range from premature infants>28 weeks to 18 years.	OMT: VO, CST, OMT techniques (ART, BLT, BLM, CS, FPR, MET, MFR or rib-raising). Treatment: median 4 (3-5)‡	UC, NT, SM, WL, SM +placebo, SM +Echinacea, postural drainage, bronchodilators.	Cerebral palsy: CHQ, GMFM-66, PEDI, WeeFIM. Respiratory: RR, EV, flow, MEP, PEF. Musculoskeletal: TM, SF, Kinesiographics (MO, MOV, MCV, OVA, CVA). Preterm infants: LOS, DWG. ADHD: Conners Scale. Infantile colic: MNHSCS. Otitis media: Antibiotic use, tympanograms, Audiometrics, SI, surgery -free months, reflectometer. CNLDO: FDT, MJT. Dysfunctional voiding: DV symptoms.	Cerebral Palsy: 6 months follow-up. Respiratory, Musculoskeletal, ADHD, congenital nasolacrimal duct obstruction, dysfunctional voiding: posttreatment. Prematurity: discharge from hospital.	No conclusive evidence on the efficacy of OMT for any paediatric condition due to i) low methodological quality of RCTs (when conditions were evaluated by individual RCTs) and ii) contradictory results for the conditions under which two RCTs were performed.	NR AEs not evaluated in 11 RCTs. No AEs occurred in 4 RCTs. 1 RCT reported patients (4) aggravation of vegetative symptoms after OMT. 1 RCT reported AEs not related to OMT.	Critically low
Neurological conditions											
Cerritelli 2017,[31] Italy	Inception to April 2016	Primary headache: migraine, tension-type headache.	5 RCTs, 235 adults. 2 migraine, 3 tension-type headache.	Gender: 78% female, 22% male (from 3 RCTs). Mean age 39.4 (from 3 RCTs).	OMT (UC, triptans, PMR): NBT (in three studies), use of protocols (in two studies). Treatment: median 4 (3-5)‡	UC, SM, OE, PMR, rest.	HIT-6 score, HF, WD, PI, DC.	Ranging from IAT to 6 months. Follow-up: 1, 3 months.	OMT reduced pain intensity, frequency and disability in patients with headache.	Number and types of AEs. AEs not evaluated in 3 RCTs, 2 RCTs reported no AEs.	Low
Visceral conditions											
Muller 2014,[32] Australia	Inception to October 2013	Irritable bowel syndrome	5 RCTs. 204 adults.	Gender: 79% female, 21% male (from 3 RCTs). Mean age 47.	OMT: applied to different body region, VO (approach on the abdomen and spine, abdomen and sacrum), NBT. Treatments: median 5 (3-5)‡	UC, SM.	Pain: VAS. Constipation, diarrhoea, AD, RS, CTT, meteorism. IBS severity score, FIS score, HAD, BDI, IBSQoL2000, FBDSI.	Ranging from 1 week to 3 months. Follow-up: short-term (2, 4 weeks), long-term (3, 12 months).	OMT, in comparison to sham therapy or standard care, reduced the symptoms of IBS, such as abdominal pain, constipation, diarrhoea, and improved general well-being.	NR All RCTs reported that no serious or statistically significant AEs occurred.	Low

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Table 1 Continued

First author, year, country	Date assessed as up to date	Conditions	Trials number, participants number	Gender distribution, age (years)	Intervention (co-intervention): description. Number of treatments (SD)	Control or comparison description	Outcomes assessed	Time points reported	Main results	Definition used to measure AEs* Reported AEs	AMSTAR-2

*Reported by the Authors of the SRs.
†In personal communications from authors of two RCTs.
‡Median (Q1–Q3).
§The number is not reported for a RCT on asthma.
¶Subgroup analysis.
**16 RCTs, only seven trials were used in our study.
IBSQoL 2000, IBS quality of life; AD, abdominal distension; ADHD, attention deficit /hyperactivity disorder; AE, adverse events; ANSLBP, acute non-specific low back pain; ART, articulatory treatment; BDI, Beck Depression Index; BLM, balanced membranous tension; BLT, balanced ligamentous tension; CHQ, child health questionnaire; CNCP, chronic non-cancer pain; CNLDO, congenital nasolacrimal duct obstruction; CNP, chronic neck pain; CNSBP, chronic non-specific body pain; CNSLBP, chronic nonspecific low back pain; CNSNP, chronic nonspecific neck pain; CP, cerebral palsy; CS, counterstrains; CST, cranial sacral therapy; CTT, colonic transit time; CV4, a technique in cranial field; CVA, cranial vault asymmetry; DC, drug consumption; DV, dysfunctional voiding; DWG, daily weight gain; EV, expiratory volume; FBDSI, functional bowel disorder severity Index; FDT, fluorescein disappearance test; FIS, fatigue impact scale; FP, frequency of pain; FPR, facilitated positional release; GMFM-66, gross motor function measure-66; HAD, hospital anxiety and depression; HF, headache frequency; HIT-6, headache impact test-6; HVLA, high velocity low amplitude thrust; IAT, immediately after treatment; IBS, irritable bowel syndrome; IC, infantile colic; JA, joint articulation; LBP, low back pain; LBP-DQ, low back pain disability questionnaire; LOS, length of stay; LT, light touch; LTP, laser therapy; MCV, maximal closing velocity; MEP, mid expiratory phase; MET, muscle energy treatment; MFR, myofascial release; MGPD, Mc Gill pain questionnaire; MJT, modified jones test; MNHSCS, mean numbers of hours spent crying and sleeping; MO, maximal mouth opening; MOV, maximal opening velocity; MRT, myofascial release treatment; MT, membranous tension; MVMA, medium velocity medium amplitude; NBT, need based treatment; NDI, Neck Disability Index; NP, not performed; NPPQ, Northwick park pain questionnaire; NPTC, not possible to calculate; NQ, Nordic questionnaire; NR, not reported; NRS, numeric rating scale; NSNP, non-specific neck pain; NT, no treatment; OCF, osteopathy in cranial field; ODI, oswestry disability Index; OE, osteopathic evaluation; OM, otitis media; OMT, osteopathic manipulative treatment; OPQ, oswestry pain questionnaire; Pedi, paediatric evaluation of disability inventory; PEF, peak expiratory flow; PGPQ, pelvic girdle pain questionnaire; PI, pain intensity; PMR, progressive muscular relaxation exercise; PP, postpartum; PT, physical therapy; QBPDS, Quebec back pain disability scale; QPP, questionnaire postpartum; QVAS, quadruple visual analogue scale; RMDQ, Roland Morris disability questionnaire; ROM, range of movement; RR, respiratory rate; RS, rectal sensitivity; SC, standard care; SD, standard deviation; SE, specific exercise; SF, spine flexibility; SI, surgical intervention; SM, sham manipulation; ST, Spencer technique; SUT, sham ultrasound treatment; SWD, short-wave diathermy; TENS, transcutaneous electrical nerve stimulation; TM, Trunk morphology; TMD, temporomandibular disorder; UC, usual care; UOBC, usual obstetrical care; VAS, visual analogic scale; VO, visceral osteopathy; WD, work disability; WeeFIM, functional independence measure for children; WL, waiting list.

Figure 1 Flow diagram of screened articles.

and may change the estimate), low quality (further research is very likely to have an important impact on confidence in the estimate of effect and is likely to change the estimate) and very low quality (there is great uncertainty about the estimate).[22] To provide a simple indication of the results for the readers, we developed a 'Traffic Light Evidence' (TLE) derived from the SR or/and MA evidence. The colour of the TLE is explained in supplemental material. Moreover, we created an 'overall TLE' (OTLE) resulting from: green light, high-quality evidence from MAs indicates intervention effectiveness; yellow light, promising evidence suggests possible effectiveness, but more research would increase our confidence in the estimate of the effect; red light, limited or inconclusive evidence.

RESULTS

Literature search results and study selection

The literature search yielded 13128 potentially relevant articles, and after eliminating duplicate articles (4778), 8350 articles were screened (see figure 1). After reading the titles and abstracts, 44 full texts were selected for eligibility of which 35 were excluded (see online supplemental table 1) and 9 SRs were considered relevant and included in this overview. A review that agreed with the outcomes of the current review was identified in PROSPERO (CRD42020162479). The authors were contacted and replied that the results were not yet available. The agreement on the eligibility of the included studies, performed by the two authors (DB and DR), resulted in a 0.78 kappa value.[23]

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Description of included reviews

This overview included nine SRs published between 2013 and 2020. Eight articles were published in English and one in Portuguese.

Six SRs focused on musculoskeletal conditions[24–29] and one each on paediatric,[30] neurological[31] and visceral conditions.[32] Detailed information on the included SRs/MAs is available in tables 1 and 2. The SRs included 71 primary studies with 5577 participants. Considering the overlapping of 14 trials and 1837 participants, the primary trials were 57 with 3740 participants (online supplemental tables 2, 3). The TLE is reported in online supplemental table 4) and the OTLE is presented in table 3 and online supplemental table 4.

Musculoskeletal conditions

Low back pain

Four reviews[24–27] with 34 RCTs (41 comparators) and 3369 participants assessed the efficacy of OMT on LBP, including acute LBP, chronic LBP (CLBP), LBP with sciatica, CLBP with menopause symptoms, LBP in obese patients, acute non-specific LBP (ANSLBP), chronic non-specific LBP (CNSLBP) and/or LBP and pelvic girdle pain in pregnancy and postpartum (PP). Taking into account overlapping, there were 22 effective trials with a total of 2053 participants.

The SR performed by De Oliveira and colleagues considered LBP in obese patients, CLBP, CLBP with sciatica and LBP in menopause or pregnancy.[24] The review included five trials with 278 participants, and three RCTs were also reported in two more SRs (see online supplemental tables 2, 3 for details). Conflicting results were derived from the primary studies. In the intergroup analysis, OMT was not effective in reducing pain in the majority of the trials. Notably, in all RCTs, the results of functional outcomes were not analysed. Using the PEDro tool, the methodological quality of the five RCTs was classified by the authors as fair to excellent (PEDro range: from 5 to 9 out of 11 points). The OTLE for OMT efficacy in reducing pain in LBP with sciatica and LBP with menopausal symptoms was assessed to be red. Adverse events were not analysed.

The SR of Franke and colleagues included 15 trials with 1502 CNSLBP or ANSLBP participants.[25] Ten trials (1141 participants) and 9 RCTs (1046 participants) investigated the effectiveness of OMT on pain and functional status, respectively. Nine RCTs were also reported in other SRs (see online supplemental tables 2, 3 for details). The meta-analysis revealed a medium and small effect in reducing pain and improving functional status, respectively, and a moderate quality of evidence (downgraded due to inconsistency). Moreover, considerable (pain) and moderate (functional status) heterogeneities were found. Similar meta-analysis results (effect and heterogeneity) have also been found in a subanalysis evaluating the effectiveness of OMT in CNSLBP patients (six trials, 771 participants). The GRADE performed by the authors revealed both a moderate quality of evidence for pain and a high quality of evidence for functional status.

Three trials (four comparators) with 242 participants evaluated the effectiveness of OMT versus obstetric care, sham ultrasound and untreated, for NSLBP in pregnant women. A large and a medium effects in reducing pain and improving functional status, respectively, were identified. Considerable (pain) and substantial (functional status) heterogeneity were found. GRADE evaluation by the authors reported a low quality of evidence for both outcomes.

Two RCTs with 119 participants evaluated the effectiveness of OMT for NSLBP in PP women. A large effect of OMT in reducing pain and improving functional status was identified. No heterogeneity was found. However, a moderate quality of evidence for both outcomes was revealed. The methodological quality of all RCTs, evaluated by the authors using the RoB from the CBRG,[18] reported a low and a high RoB for 13 and 2 RCTs, respectively. However, considering the last version of the CBN Group,[19] we rated all RCTs at high RoB [domains at high RoB (% of RCTs): care provider (100%), patient blinding (67%), outcome assessor blinding (67%), groups similar at baseline (27%), lack of intention to treat analysis use (27%), free of selective outcome report (13%), dropouts described+acceptable (7%), similar timing outcome assessment (7%) and compliance acceptable (7%)]. The OTLE for the outcomes of each condition was assessed to be yellow.

Adverse events were evaluated in only 4 out of the 15 primary studies. Two RCTs reported minor adverse events such as stiffness and tiredness, one no adverse event and the last one evidenced adverse events not related to the treatment intervention.

In another SR, Franke and colleagues[26] identified eight RCTs with 850 participants evaluating the efficacy of OMT on NSLBP and pelvic girdle pain in pregnancy (five RCTs, seven comparisons) and on NSLBP in PP women (three trials and three comparisons) (see online supplemental tables 2, 3 for overlapping). The pooled analysis of five RCTs with 677 pregnancy participants reported the efficacy of OMT in reducing pain and improving functional status; however, a medium effect and a considerable heterogeneity were revealed. The GRADE performed by the authors indicated a moderate quality of evidence.

The meta-analysis including three studies with 173 PP participants revealed a significant effect in favour of OMT in reducing pain and improving functional status, although a large effect and substantial/considerable heterogeneity for both outcomes were reported. The GRADE performed by the authors also found a low quality of evidence.

The methodological quality of the included studies evaluated by the authors using the CBRG,[18] identified a low RoB for all RCTs. Considering the CBN Group,[19] we rated all RCTs as at high RoB [domains at high RoB (% of RCTs): patient binding (100%), care provider binding (100%), outcome assessor blinding (100%), dropouts

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Table 2 Quality of the primary RCTs included in the systematic reviews/meta-analyses and meta-analyses quantitative results

First author, year, country	Primary studies quality. GRADE	Meta-analysis data
Musculoskeletal conditions: low back pain		
De Oliveira Meirelles, 2013,[24] Brazil	Pedro score: 6 (2 RCTs), 9 (1 RCT), 7 (1 RCT), 5 (1 RCT).	NP
Franke, 2014,[25] Australia	Low RoB (13 RCTs, low risk of bias in at least six categories). High RoB (2 RCTs).	
	GRADE	
	ANSLBP and CNSLBP	
	Pain: MODERATE	Pain: (MD −12.91; 95% CI: −20.00 to −5.82). I^2=86%.
	Functional status: MODERATE	Functional status: (SMD −0.36; 95% CI: −0.58 to −0.14). I^2=57%.
	CNSLBP	
	Pain: MODERATE	Pain: (MD −14.93; 95% CI: −25.18 to −4.68). I^2=89%.
	Functional status: HIGH	Functional status: (SMD −0.32; 95% CI: −0.58 to −0.07). I^2=49%.
	NSLBP in pregnancy	
	Pain: LOW	Pain: (MD −23.01; 95% CI: −44.13 to −1.88). I^2=91%.
	Functional status: LOW	Functional status: (SMD −0.80; 95% CI: −1.36 to −0.23). I^2=76%.
	NSLBP in PP	
	Pain: MODERATE	Pain: (MD −41.85; 95% CI: −49.43 to −34.27). I^2=0%.
	Functional status: MODERATE	Functional status: (SMD −1.78; 95% CI: −2.21 to −1.35). I^2=0%.
Franke, 2017,[26] Australia	Low RoB (all RCTs, low risk of bias in at least six categories).	
	GRADE	
	NSLBP in pregnancy	
	Pain: MODERATE	Pain: (MD −16.75; 95% CI: −31.79 to −1.72). I^2=94%.
	Functional status: MODERATE	Functional status: (SMD −0.50; 95% CI: −0.93 to −0.07). I^2=84%.
	LBP in PP	
	Pain: LOW	Pain: (MD −38.00; 95% CI: −46.75 to −29.24). I^2=68%.
	Functional status: LOW	Functional status: (SMD −2.12; 95% CI: −3.02 to −1.22). I^2=81%.
Dal Farra, 2020,[27] Italy	High RoB (all RCTs).	
	GRADE	
	CNSLBP	
	Pain: LOW	Pain: (SMD −0.57; 95% CI: −0.90 to −0.25). I^2=72%.
	Functional status: LOW	Functional status: (SMD −0.34; 95% CI: − 0.65 to −0.03). I^2=71%.
	Functional status (12 weeks follow-up): LOW	Functional status 12 weeks follow-up: (SMD −0.14; 95% CI: −0.31 to 0.03). I^2=0%.
Musculoskeletal conditions: neck pain		
Franke, 2015,[28] Australia	Low RoB (all RCTs, low risk of bias in at least six categories).	
	GRADE	
	CNSNP	
	Pain: MODERATE	Pain: (MD −13.04, 95% CI: −20.64 to −5.44). I^2=34%.
	Functional status: MODERATE	Functional status: (SMD: −0.38, 95% CI: −0.88 to 0.11). I^2=0%.

Continued

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Table 2 Continued

First author, year, country	Primary studies quality. GRADE	Meta-analysis data
Musculoskeletal conditions: chronic non-cancer pain		
Rehman, 2020,[29] Canada	High RoB (all RCTs, based on a modified RoB with six domains).	
	GRADE	
	CNCP	
	Pain: MODERATE	Pain (OMT vs SC): (SMD – 0.37; 95% CI: – 0.58 to –0.17). I^2=25%.
	Disability: MODERATE	Disability (OMT vs SC): (SMD –1.04; 95% CI: – 1.23 to –0.85). I^2=0%.
	Quality of life: MODERATE	Quality of life (OMT vs SC): (SMD 0.67; 95% CI: 0.29 to 1.05). I^2=0%.
Paediatric conditions		
Posadzki, 2013,[30] South Korea	High risk (all RCTs).	NP
Neurology conditions		
Cerritelli, 2017,[31] Italy	JADAD NR*. The majority of RCTs have high or unclear RoB.	NP
Visceral conditions		
Muller, 2014,[32] Australia	Low RoB (all RCTs, low risk of bias in at least six categories).	NP

*Reported in methods but not performed.
ANSLBP, acute non-specific low back pain; CNCP, chronic non-cancer pain; CNP, chronic neck pain; CNSBP, chronic non-specific body pain; CNSLBP, chronic non- specific low back pain; CNSNP, chronic non-specific neck pain; MD, mean difference; NP, not performed; NR, not reported; OMT, osteopathic manipulative treatment; PP, postpartum; RCT, randomised controlled trial; RoB, risk of bias; SC, standard care; SMD, standard mean difference.

described +acceptable (25%), group similar at the baseline (25%), intention to treat analysis (25%), similar timing outcome assessment (25%) and compliance acceptable (12%)]. The OTLE for the outcomes of each condition was assessed to be yellow.

Concerning the adverse events, one study reported occasional tiredness in some patients after OMT, two studies (personal communications to authors SR) did not find adverse events and the remaining five studies did not analyse adverse events.

The SR by Dal Farra and colleagues[27] evaluated the effectiveness of osteopathic interventions, performed by any type of manual therapist in CNSLBP patients. A subgroup analysis evaluating the effectiveness of OMT performed only by osteopaths identified six trials (eight comparisons) with 739 participants; five trials were also reported in other two further SRs (see online supplemental tables 2, 3 for more details).

The authors revealed a significant effect, clinically relevant according to the CBN Group,[19] of OMT in reducing pain (medium effect) and improving functional status (small effect). However, substantial heterogeneity and a low quality of evidence (GRADE) were reported for both outcomes.

A further subanalysis, including two trials (three comparisons) with 548 participants, did not find evidence of OMT efficacy on functional status after a long-term treatment (12 weeks follow-up). Low quality of evidence and no heterogeneity were reported. The methodological quality of the primary studies, evaluated by the authors using the CBN Group,[19] reported a high RoB for all RCTs [domains at high RoB (% of RCTs): high RoB for care provider (100%), patient blinding (50%), outcome assessor blinding (17%), participant allocation (33%) and reporting bias (17%)]. The OTLE for the outcomes was assessed to be yellow.

With regards to adverse events, a trial reported an increase in back muscle spasticity in one patient treated with OMT.

Neck pain

Franke and colleagues[28] evaluating 3 RCTs (three comparators) with 123 participants, provided evidence that OMT exerted beneficial effects on chronic non-specific neck pain (CNSNP). Specifically, a medium ES in reducing pain and a moderate quality of evidence on pain outcome were reported. A low level of heterogeneity was found. However, the meta-analysis did not show a significant effect on functional status. The methodological quality of all RCTs, evaluated by the authors using the CBRG,[18] reported a low RoB for all RCTs. Considering the CBN Group,[19] we rated all RCTs at high RoB [domains at high RoB (% of RCTs): patient blinding (67%), care provider (100%), outcome assessor blinding (67%), dropouts described+acceptable (33%) and intention to

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Table 3 Overall traffic light evidence for OMT efficacy

Conditions	First author, year	Overall traffic light evidence
Musculoskeletal conditions		
1.ANSLBP/CNSLBP		
Pain	Franke, 2014[25]	
	Dal Farra, 2020[27]	
Functional status	Franke, 2014[25]	
	Dal Farra, 2020[27]	
2.CNSLBP		
Pain	Franke, 2014[25]	
	Dal Farra, 2020[27]	
Functional status	Franke, 2014[25]	
	Dal Farra, 2020[27]	
3.NSLBP in pregnancy		
Pain	Franke, 2014[25]	
	Franke, 2017[26]	
Functional status	Franke, 2014[25]	
	Franke, 2017[26]	
4.NSLBP in PP		
Pain	Franke, 2014[25]	
	Franke, 2017[26]	
Functional status	Franke, 2014[25]	
	Franke, 2017[26]	
5.LBP with sciatica		
Pain	De Oliveira Meirelles, 2013[24]	
6.LBP with menopausal symptoms		
Pain	De Oliveira Meirelles, 2013[24]	
7.CNSNP		
Pain	Franke, 2015[28]	
Functional status	Franke, 2015[28]	
8.CNCP		
Pain	Rehmanl, 2020[29]	
Disabiliy	Rehman, 2020[29]	
Quality of life	Rehman, 2020[29]	
Paediatric conditions		
Outcomes for different conditions *	Posadzky, 2013[30]	
Neurological conditions		
Outcomes for migraine and tension-type headache†	Cerritelli, 2017[31]	
Visceral conditions		
Outcomes for IBS‡	Muller, 2014[32]	

Overall traffic light evidence: yellow light, promising evidence suggests possible effectiveness, but more research would increase our confidence in the estimate of the effect; red light, limited or inconclusive evidence.
*Different conditions were considered. It is not possible to evaluate the single outcome for each condition.
†Pain, work disability, headache frequency, quality of life.
‡Pain, constipation, quality of life.
ANSLBP, acute non-specific low back pain; CNCP, chronic non-cancer pain; CNSLBP, chronic non- specific low back pain; CNSNP, chronic non-specific neck pain; IBS, irritable bowel syndrome; LBP, low back pain; NSLBP, non-specific low back pain; PP, postpartum.

treat analysis (100%)]. The OTLE for the outcomes was assessed to be yellow.

No serious adverse events occurred in any RCTs (data reported in an RCT and as personal communications to SR authors in the other two studies).

Chronic non-cancer pain

The SR by Rehman and colleagues[29] evaluated the efficacy of osteopathic interventions performed by manual therapists in chronic non-cancer pain. Sixteen RCTs were identified; however, we considered pooled analyses in which the trials were only performed by osteopaths (see online supplemental tables 2, 3 for overlapping). A pooled analysis, including 6 RCTs with 728 participants (six comparators), found the efficacy of OMT versus standard care in reducing pain severity (small ES, moderate quality of evidence and low level of heterogeneity). Moreover, another pooled analysis including two trials with 486 participants revealed the efficacy of OMT versus standard care in improving disability (large ES, moderate quality of evidence and no heterogeneity). Similarly, the pooled analysis of the other two trials with 210 participants found that OMT versus standard care improved the quality of life (medium effect, moderate quality of evidence and no heterogeneity).

The methodological quality of the included studies was performed by the authors using a modified version of the Handbook of Cochrane[33] where only six domains were considered (random sequence generation, allocation concealment, blinding of participants, healthcare provider, outcome assessors and dropout rates). According to this modified version, the quality of the RCTs was reported by the authors to be at high RoB [domains at high RoB (% of RCTs): for patient blinding (100%), care provider blinding (100%), outcome assessor blinding (57%), random sequence generation (29%), participant allocation concealment (29%), and dropout >20% (43%)]. The OTLE for the outcomes of each condition was assessed to be yellow.

Adverse events were not considered by the SR authors.

Paediatric conditions

The SR by Posadzky and colleagues[30] evaluated the efficacy of OMT in paediatric conditions. This review included 17 RCTs involving a total of 887 participants with different conditions: cerebral palsy evaluated in two clinical studies involving a total of 197 participants, respiratory conditions evaluated in four trials involving 186 patients [obstructive apnoea one RCT, asthma two RCTs (in one study not reported the number of patients), bronchiolitis one RCT], otitis media evaluated in three trials involving a total of 167 participants, musculoskeletal function evaluated in three trials with 80 patients (idiopathic scoliosis one RCT, mandibular kinematics one RCT, postural asymmetry one RCT) and attention-deficit/hyperactivity disorder (77 participants), prematurity (101 participants), infantile colic (28 participants), congenital nasolacrimal duct obstruction (30 patients)

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and functional voiding (21 participants) individually assessed by one RCT. The single trials provided evidence that OMT exerted beneficial effects on congenital nasolacrimal duct obstruction (post-treatment), daily weight gain and length of hospital stay, dysfunctional voiding, infantile colic and postural asymmetry. By contrast, no significant effects of OMT on idiopathic scoliosis, obstructive apnoea or temporomandibular disorders compared with various control interventions have been evidenced by the single RCTs. For conditions in which more than one RCT has been performed (asthma, otitis media and cerebral palsy), contradictory results are reported. From the SR, it emerges that low-quality RCTs favoured OMT, while moderate and high-quality RCTs failed to find OMT effectiveness. The vast majority of the RCTs were reported by the authors to be at high RoB (15 RCTs) [domains at high RoB (% of RCTs): allocation concealment (67%), patient blinding (67%), care provider blinding (100%), outcome assessor blinding (50%), addressing of incomplete data (33%), selective outcome reporting (33%), adequate sequence generation (28%)] with unclear or low RoB for the remaining two RCTs. The OTLE for outcomes of each condition was assessed to be red.

In 11 RCTs, adverse events were not analysed. No adverse events or serious adverse events following OMT were reported in four trials. Adverse events occurred in one RCT, but they were not related to OMT. One trial reported the aggravation of vegetative symptoms in four patients.

Neurological conditions

The SR of Cerritelli and colleagues,[31] including five RCTs with a total of 235 participants, evaluated two different types of primary headache: migraine (two RCTs, 147 participants) and tension-type headache (three RCTs, 88 participants). Although the two RCTs evaluating efficacy in migraine reported positive results in favour of OMT (mainly in pain intensity reduction), intergroup analysis was performed only in one RCT. Similarly, evidence has been reported for the tension-type headache only when within-group analysis was performed; intergroup analyses reported conflicting results. The RCTs were reported by the authors to be at high RoB [domains at high RoB (% of RCTs): care provider blinding (100%), participant blinding (60%) and allocation concealment (20%)]. Due to high heterogeneity (different types of primary headaches, different outcome measures and variable length of follow-up), a meta-analysis was not conducted by the authors. The OTLE for the outcomes of each condition was assessed to be red.

Adverse events, evaluated in two RCTs, did not occur.

Visceral conditions

In a SR, Muller and colleagues,[32] including five primary studies and involving 204 participants, evaluated the efficacy of OMT in the treatment of irritable bowel syndrome (IBS). Although high heterogeneity (in outcome measures and follow-up period) was evidenced, the results indicated that OMT was effective in IBS. The methodological quality of all RCTs, evaluated by the authors using the CBRG,[18] reported a low RoB for all RCTs. Considering the CBN Group,[19] we rated all RCTs at high RoB [domains at high RoB (% of RCTs): care provider blinding (100%), outcome assessor blinding (60%), randomisation (20%), patient blinding (20%), groups similar at the baseline (20%) and intention to treat analysis (20%)]. The OTLE for the outcomes was assessed to be red.

No adverse events occurred in the patients from any of the RCTs.

Methodological quality of included reviews

A summary of the findings of the AMSTAR-2 tool is provided in tables 1 and 4. Inter-rater agreement between the two overview authors (DB and FB) on the ranking of quality achieved a 0.89 kappa value.[23]

According to the critical domain established in Shea *et al*,[12] seven[25–29 31 32] and two SRs [24 30] were rated as low and critically low quality, respectively.

Two of the nine SRs registered a protocol before beginning the study.[27 29] Eight SRs performed an appropriate literature search,[25–32] and five SRs reported justification for the exclusion of primary studies.[25 26 28 31 32] All SRs[24–32] evaluated the RoB of the included studies and five SRs[25–29] carried out a meta-analysis and used appropriate methods for the statistical combination of findings. Eight SRs[25–32] accounted for the RoB when interpreting and discussing the results of the SR. Finally, domain 15 (publication bias assessment) was rated as not applicable for all the SRs due to lack of a meta-analysis[24 30–32] or the inclusion in the meta-analysis of fewer than 10 trials.[25–29]

DISCUSSION

Osteopathic medicine, a form of complementary medicine, is a type of manual therapy used to normalise the structure–function relationship and to promote the body's own self-healing mechanism. In the last decade, CAM therapies have grown in use and popularity, and among these, many surveys have demonstrated an increasing interest in osteopathic medicine in patients with musculoskeletal disorders such as CNSLBP and neck pain.[34 35]

Recently, osteopathic medicine has been regulated in many countries including the USA, Australia, the UK, Iceland, Denmark, Malta, Portugal and Switzerland, where it is a primary healthcare profession. In other countries, the regulation process has not yet been completed (ie, Italy), or there is no legal recognition of the osteopathic profession.[36] In this context, we performed an overview to summarise the best available clinical evidence on the efficacy and safety of OMT. We identified nine SRs on the use of osteopathic care for the management of musculoskeletal, paediatric, visceral and neurological disorders with different effects and clinical relevance depending on the conditions.

From our overview, some relevant questionable problems emerge related to the lack of appropriate guidelines for assessing the methodological quality of trials in manual therapy and problems due to inadequate reporting of trial methodology and results. In this regard, most of the trials

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Table 4 Quality assessment of the included systematic reviews by the AMSTAR-2 tool

First author, year	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9 RCT	Q9 NRSI	Q10	Q11 RCT	Q11 NRSI	Q12	Q13	Q14	Q15	Q16	Ranking of quality
Musculoskeletal conditions																			
De Oliveira Meirelles, 2013[24]	N	N	N	N	N	N	N	PY	Y	N/A	N	N/A	N/A	N/A	N	N	N/A	N	CRITICALLY LOW
Franke, 2014[25]	Y	N	N	Y	Y	Y	Y	PY	Y	N/A	N	Y	N/A	Y	Y	Y	N/A	Y	LOW
Franke, 2017[26]	Y	N	N	Y	Y	Y	Y	PY	Y	N/A	N	Y	N/A	Y	Y	Y	N/A	N	LOW
Dal Farra, 2020[27]	Y	Y	Y	Y	Y	Y	N	PY	Y	N/A	N	Y	N/A	Y	Y	Y	N/A	Y	LOW
Franke, 2015[28]	Y	N	N	Y	Y	Y	Y	PY	Y	N/A	N	Y	N/A	Y	Y	Y	N/A	Y	LOW
Rehman, 2020[29]	Y	Y	N	Y	Y	Y	N	PY	Y	N/A	N	Y	N/A	Y	Y	Y	N/A	Y	LOW
Paediatric conditions																			
Posadzki, 2013[30]	Y	N	N	PY	Y	Y	N	PY	Y	N/A	Y	N/A	N/A	N/A	Y	Y	N/A	Y	CRITICALLY LOW
Neurology conditions																			
Cerritelli, 2017[31]	Y	N	N	Y	Y	Y	Y	Y	Y	N/A	Y	N/A	N/A	N/A	Y	Y	N/A	Y	LOW
Visceral conditions																			
Muller, 2014[32]	Y	N	N	PY	Y	Y	Y	PY	Y	N/A	N	N/A	N/A	N/A	Y	N	N/A	Y	LOW

N, no; N/A, not applicable; NRSI, non-randomised studies of interventions; PY, partial yes; RCT, randomised controlled trial; Y, yes.

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included in the SRs reported a high or unclear RoB for blinding procedures: patient, outcome assessor and care provider blinding. In manual therapy, blinding is an issue, as patients tend to be aware of the manual treatment and therapists cannot be blinded to the treatment intervention they deliver.[37] For participant-reported outcomes, for which the patient is the outcome assessor, such as for pain and functional status outcomes, blinding of patients is mandatory, and, therefore, it is necessary to use, as a control group, sham procedures (including light touch therapy) that simulate manipulation. These sham procedures should be reported in RCTs; however, a lack of reporting placebo sham therapy procedures in both SRs and primary studies has been evidenced. It is important to note that, although these findings have already been reported by Cerritelli and colleagues,[38] to date, these suggestions have not been followed. More effort should be made to promote guidelines for designing the most reliable placebo for manual treatment to reduce the RoB in patient blinding. However, a recent meta-epidemiological study found no evidence that lack of patients' blinding had an impact on estimate effects.[39]

Other issues that emerge from our overview are the lack of treatment description and timing of measuring outcomes (short term and long term) in the SRs as well as in primary trials. In osteopathic medicine, as in any other manual therapy, it is important to describe in adequate detail each phase of the intervention, including how and when they were administered, and when the outcomes are measured. Without a complete description of treatments, clinicians cannot reliably reproduce useful interventions. Proper checklists for non-pharmacological treatments, such as the Template for Intervention Description and Replication guide/checklist and the CONsolidated Standards of Reporting Trials statement for randomised non-pharmacological treatment studies, should be followed by clinical trial authors.[40 41]

That said, our overview highlights that evidence on the efficacy of OMT is: (1) promising in musculoskeletal disorders, mainly in reducing pain and improving functional status in acute and chronic NSLBP patients, NSLBP in pregnancy or PP (OTLE: yellow), (2) limited and contradictory in the treatment of paediatric conditions (some conditions were evaluated by only one trial, some of which were of low methodological quality, and contradictory results were obtained for conditions in which two RCTs were performed, OTLE: red) and (3) limited on primary headache and IBS (OTLE: red).

The lack of solid evidence stems from a small sample size,[26 28–32] the presence of conflicting results[24 30 31] and a high heterogeneity in participants,[25 31] outcomes measures,[31 32] interventions[25–27 31] and comparison interventions.[25–27 32] Notably, reduced heterogeneity was found when the RCTs were pooled considering interventions and comparators.[29]

According to AMSTAR-2, the methodological quality of the included SRs was rated low and critically low. Domain 2 (registered protocol) was critical for seven SRs. The presence of a written and registered protocol prior to conducting the review should ensure that review methods are transparent and reproducible, and adhere to this prespecified research plan.[42] These should help avoid bias and unintended duplication of reviews.

Adverse events

In general, manual therapies have been reported to be well tolerated, and manual therapy-related adverse events are short lived and mild or moderate in intensity.[43] In our overview, we found that seven SRs[25–28 30–32] evaluated adverse events, and from these SRs, it emerges that no severe incident involving musculoskeletal, neurological, visceral or paediatric disorders occurred after OMT. However, it should be noted that among these seven SRs, only two reported the definition used to measure adverse events. The idea that manual therapies are safe could only be demonstrated if adverse events are defined and assessed in each clinical trial. Specifically, the authors should adequately report in detail the approach used to measure adverse events, which need to be defined using an appropriate taxonomy.[44 45]

Strengths and limitations

Numerous limitations can be found in our overview. First, considering our inclusion criteria, we may have missed some relevant SRs. Indeed, we included SRs by evaluating only RCTs (and not other study designs), in which OMT was performed by osteopathic physicians or osteopaths (and not by other manual therapists). Globally, two professional figures have emerged, largely due to different legal and regulatory systems around the world: osteopathic physicians, who are doctors with full and unlimited medical practice rights, and osteopaths who have obtained academic and professional standards for diagnosing and practicing treatments based on the principles of osteopathic philosophy. OMT is the core activity for both osteopathic physicians and osteopaths who follow the principles of osteopathic medicine by performing a personalised treatment according to the patient evaluation and subsequent tailoring.[46] Therefore, our decision to consider only osteopathic physicians or osteopaths arises from the premise of avoiding the fact that the principles of osteopathic medicine are not followed. In this regard, we excluded seven SRs, and, therefore, considering the overlapping, five RCTs were lost (see online supplemental table 1 for details). According to our decision, a recent scoping review used more restrictive inclusion criteria, considering only studies performed in the USA where OMT is practiced by osteopathic physicians.[47] Considering that in most countries osteopathy is often conducted in the private sector (eg, the UK, France and Italy), the participants included in the primary studies might not be generalisable to the population.

Since RCTs are widely recognised as the best design for evaluating the efficacy of an intervention, we also decided to include only SRs evaluating RCTs. In this regard, 11 systematic reviews were excluded and, considering the overlapping, 17 RCTs were lost (see online supplemental table 1 for details).

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CONCLUSION

This overview suggests that OMT could be effective in the management of musculoskeletal disorders, specifically with regard to CNSLBP patients and LBP in pregnant or PP women. In contrast, inconclusive evidence was derived from SRs analysing the OMT efficacy on paediatric conditions, primary headache and IBS. Although not all RCTs have investigated the safety of OMT, considering that no serious adverse events have been reported, OMT can be considered safe.

Nevertheless, based on the low number of studies, some of which are of moderate quality, our overview highlights the need to perform further well-conducted SRs as well as clinical trials (which have to follow the specific guidelines for non-pharmacological treatments) to confirm and extend the possible use of OMT in some conditions as well as its safety.

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